General relativity HW-9

Problem 1. Consider FLRW (or Friedmann) Universe with matter. Consider cases of zero spatial curvatures. Consider separately two kinds of matter: dust and radiation. For each of this matter components derive a time dependence of a scale-factor.

Solution:
By Friedmann equation

$$(\frac{\dot{a}}{a})^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}$$

Because we are considering zero spatial curvature,then $k = 0$.
For dust,$\rho = ma^{-3}$,then

$$a^{\frac{1}{2}}da = \frac{8\pi Gm}{3}dt \implies a^{\frac{3}{2}} - a_0^{\frac{3}{2}} = 4\pi Gmt \implies a \propto t^{\frac{2}{3}}$$

For radiation,$\rho = ma^{-4}$,then

$$ada = \frac{8\pi Gm}{3}dt \implies a^2 - a_0^2 = \frac{16\pi Gm}{3}t \implies a \propto t^{\frac{1}{2}}$$

Problem 2. The same, but take the spatial curvature to be ± 1 and derive answers for a radiation only.

Solution:
For $k = 1$,

$$(\frac{\dot{a}}{a})^2 = \frac{8\pi Gm}{3a^4} - \frac{1}{a^2} \implies \dot{a}^2 = \frac{\beta^2}{a^2} - 1 \implies \int \frac{ada}{\sqrt{\beta^2 - a^2}} = \int dt$$

Where $\beta = \sqrt{\frac{8\pi Gm}{3}}$.Let $a = \beta sin\gamma$,then if $a_0 = 0$,

$$\int \frac{ada}{\sqrt{\beta^2 - a^2}} = \int \frac{\beta^2 sin\gamma cos\gamma d\gamma}{\beta cos\gamma} = \sqrt{\beta^2 - a_0^2} - \sqrt{\beta^2 - a_1^2} \implies a^2 = 2\beta t - t^2$$

Clearly $t \in [0, 2\beta]$ and $a \in [0, \beta^2]$.
For $k = -1$,

$$(\frac{\dot{a}}{a})^2 = \frac{8\pi Gm}{3a^4} + \frac{1}{a^2} \implies \dot{a}^2 = \frac{\beta}{a^2} + 1 \implies \int \frac{ada}{\sqrt{\beta^2 + a^2}} = \int dt$$

Let $a = \beta tan\gamma$,then if $a_0 = 0$,

$$\int \frac{ada}{\sqrt{\beta^2 + a^2}} = \int \frac{\beta sin\gamma d\gamma}{cos^2\gamma} = \frac{\beta}{cos\gamma} = \sqrt{\beta^2 + a_1^2} - \sqrt{\beta^2 + a_0^2} \implies a^2 = t^2 + 2\beta t$$

For large t,$a \sim t$.This agrees with the figure in Carroll's book,where $k = +1$ is a closed curve,and $k = -1$ increase faster than $k = 0$.